

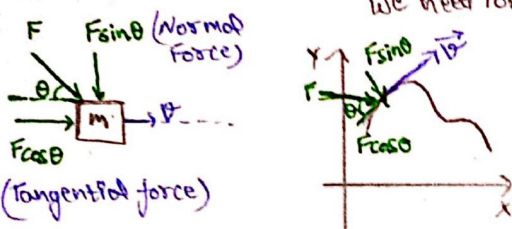
Classification of forces:

	strength
1) Gravitational	→ Appearance
2) Weak Force	→ $\times 10^{25}$
3) EM Force	→ $\times 10^{36}$
4) Strong Force	→ $\times 10^{38}$

Force and state of Motion:

→ State of a motion is defined by its velocity vector.

$\vec{v}$ → Magnitude
→ Direction
To change both of these parameters we need Force.



→ Normal Force

① Responsible for change in dirⁿ of velocity.

$$a_N = \frac{F \sin \theta}{m} = \frac{v^2}{R}$$

(Also called Centripetal Accn)

→ Tangential Force

① Responsible for change in magnitude of velocity.

$$a_t = \frac{dv}{dt} = \frac{F \cos \theta}{m}$$

→ Total Acceleration

The net acceleration produced by force F can be given by

$$a_t = \frac{dv}{dt}$$

$$a_r = \sqrt{a_t^2 + a_N^2}$$

Static Equilibrium:

At any kind of equilibrium,

$$\sum \vec{F} = 0 \text{ (Balanced Forces)}$$

$$\Rightarrow \vec{a}_{\text{net}} = 0$$

Newton's Laws of Motion

$$\Rightarrow \vec{v} = \text{constant}$$

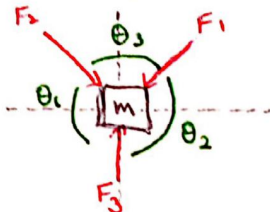
$$\vec{v} = 0 \text{ (Rest)}$$

$$|\vec{v}| \neq 0$$

This condⁿ is static eq^m.

Lami's Theorem:

→ When static eq^m is maintained due to three forces, then acc^t to this theorem they behave like vertices of a triangle.



→ Then by sine law,

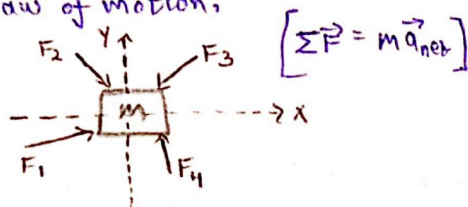
$$\frac{F_1}{\sin \theta_1} = \frac{F_2}{\sin \theta_2} = \frac{F_3}{\sin \theta_3}$$

→ In case of more than 3 forces, resolve them along x & y, and

$$\left[\sum \vec{F}_x = 0 \quad \sum \vec{F}_y = 0 \right]$$

Unbalanced forces:

→ We deal such cases with second law of motion,



$$\sum F_x = m a_x$$

$$\sum F_y = m a_y$$

$$a_{\text{net}} = \sqrt{a_x^2 + a_y^2}$$

Eg:



$$a_x = a_1 - a_2 \cos \theta$$

$$a_y = a_2 \sin \theta$$

→ For a system of particles,

$$\sum \vec{F} = \sum m \cdot a_{\text{com}}$$

Action-Reaction of Common Forces:

→ Action-Reaction forces are of same magnitude but opp in dirⁿ.

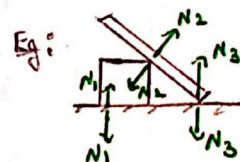
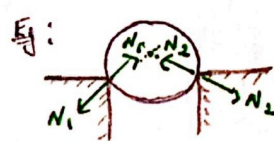
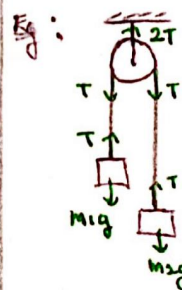
→ Action-Reaction forces acts on different surfaces.

External Forces → mg , F_{ext} .

Internal Forces → T , N and f
Friction

→ Normal contact force always acts $\perp$ to contact surface.

→ Tension at every point of the same massless string is same.



Constrained Motion:

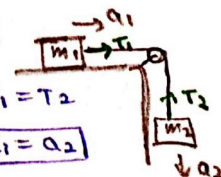
$$a_{\text{net}} = \frac{\sum F_{\text{along } a} - \sum F_{\text{opp } a}}{\sum m}$$

[Neglect the forces $\perp$ to 'a']

→ Since tension is an internal force, so the work done by it on system is zero.

$$\therefore \text{ (Positive Work)}_T = \text{ (Negative Work)}_T$$

Eg:



$$T_1 = T_2$$

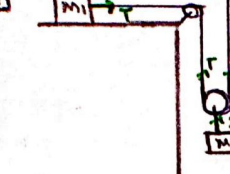
$$a_1 = a_2$$

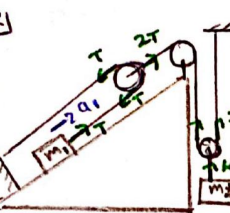
$$T_1 x_1 = T_2 x_2$$

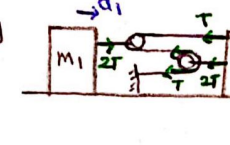
$$T_1 v_1 = T_2 v_2$$

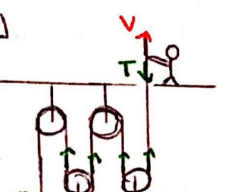
$$T_1 a_1 = T_2 a_2$$

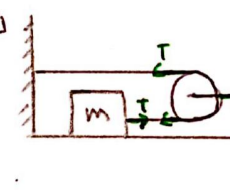
Most important to get relation b/w accⁿs. Simply write these eq^s for given systems

1)  $T a_1 = 2T \cdot a_2$
 $\Rightarrow a_1 = 2a_2$

2)  $T \cdot a_1 = 4T \cdot a_2$
 $\Rightarrow a_1 = 4a_2$

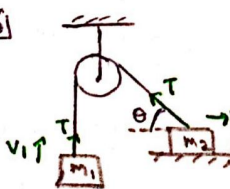
3)  $2T \cdot a_1 = 3T \cdot a_2$
 $2a_1 = 3a_2$

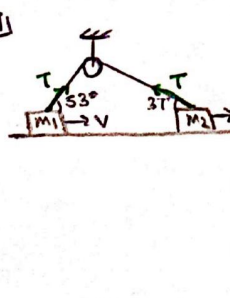
4)  Find velocity of pulley.
Solⁿ $5T \cdot v_p = T \cdot v$
 $\Rightarrow v_p = v/5$

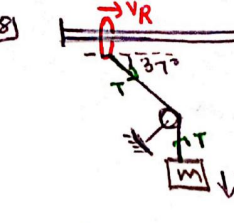
5)  Find accⁿ of ideal pulley.
Solⁿ $2T = F$
 (Due to mass = 0
 Not because of eq^m)

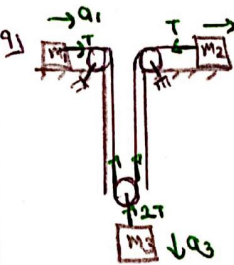
Also, $T = m \cdot a_{\text{Block}} \Rightarrow a_{\text{Block}} = \frac{F}{2m}$

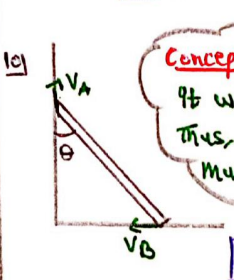
Now, $a_{\text{pulley}} \cdot 2T = a_{\text{Block}} \cdot T$
 $\Rightarrow a_{\text{pulley}} = \frac{a_{\text{Block}}}{2} = \frac{F}{4m}$

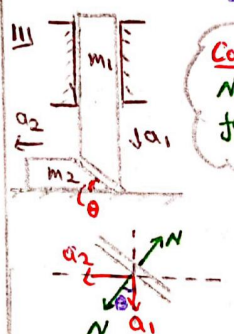
6)  $T \cdot v_1 = T \cdot \cos \theta \cdot v_2$
 $\Rightarrow v_1 = v_2 \cos \theta$

7)  Find velocity of m_1 .
Solⁿ $T \cos 53^\circ \cdot v = T \cos 37^\circ \cdot (10)$
 $\Rightarrow \frac{3}{5} v = \frac{4}{5} \times 10$
 $\Rightarrow v = \frac{40}{3} \text{ m/s}$

8)  Find velocity of Ring.
Solⁿ $V_R \cdot T \cos 37^\circ = V_T$
 $\Rightarrow V_R = \frac{5V}{3}$

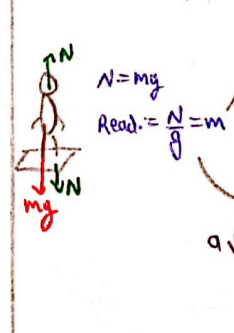
9)  $T a_1 = 2T a_2 + T a_3$
 $\Rightarrow a_3 = \frac{a_1 - a_2}{2}$

10)  Concept: Since rod is rigid, it won't compress/elongate. Thus, velocity along length must be same.
 $V_A \cos \theta = V_B \sin \theta$

11)  Concept: Just like tension, Normal would be int. force. Make its work done zero.
 $N \sin \theta \cdot a_2 = N \cos \theta \cdot a_1$
 $\Rightarrow a_1 = a_2 \tan \theta$

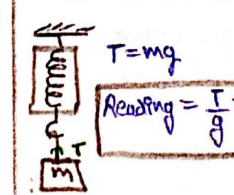
Concept of Weighing Machine :

→ Measure Normal rxⁿ force acting on the surface.

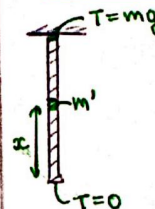
 $N = mg$
 Read. = $\frac{N}{g} = m$
 $N - mg = ma$
 $N = m(g + a)$
 Read. = $m(1 + \frac{a}{g})$
 $mg - N = ma$
 $N = m(g - a)$
 Read. = $m(1 - \frac{a}{g})$

Concept of Spring Balance :

→ Measures tension in string attached to it.

 $T = mg$
 Reading = $\frac{T}{g} = m$
 $T = \frac{200}{3} \text{ N} \Rightarrow \text{Read.} = \frac{20}{3} \text{ N}$

Tension in a Rope :

 Tension at all points of a rope of mass m is different.
 $T' = m'g = \left(\frac{mx}{L}\right)g$

Variable Force :

 $F(t) = m \cdot \frac{dv}{dt} \Rightarrow \int_0^t F(t) dt = \int_0^v m \cdot dv$
 $F(x) = m \cdot \frac{dv}{dx} \Rightarrow \int_0^x F(x) dx = \int_0^v m \cdot dv$
 $F(v) = m \cdot \frac{dv}{dt} \Rightarrow \int_0^v F(v) dv = \int_0^t m \cdot dt$

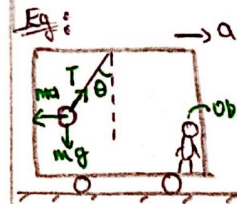
Pseudo Force :

→ Acts when NLM applied from a non-inertial frame.

On which? Body on which we apply NLM

Magnitude? (Mass of Body) × (Accⁿ of frame w.r.t ground)

Direction? Opp. to dirⁿ of accⁿ of frame

Ex:  → To observer, bob of mass m appears at rest. True

$T \sin \theta = ma$
 $T \cos \theta = mg$
 $\tan \theta = a/g$
 $a = g \tan \theta$
 $T = m \sqrt{a^2 + g^2}$